Newman-Penrose Formalism Calculations

Stephani et al., Equation (33.40)

Metric

$$ds^2 = -\Phi^2 \mathrm{d}t \otimes \mathrm{d}t + \left(\frac{\Phi^2 \left(\frac{\partial P}{\partial r}\right)^2 - 2\,P\Phi \frac{\partial P}{\partial r} \frac{\partial \Phi}{\partial r} + P^2 \frac{\partial \Phi}{\partial r}^2}{P^2} \right) \mathrm{d}r \otimes \mathrm{d}r + \frac{\Phi^2}{P^2} \mathrm{d}x \otimes \mathrm{d}x + \frac{\Phi^2}{P^2} \mathrm{d}y \otimes \mathrm{d}y$$

Coordinates

$(x^\mu) = (t, r, x, y)$

Metric Functions

$$P = x^2 \alpha\,(r) + y^2 \alpha\,(r) + x\beta\,(r) + i\,y\beta\,(r) + x\overline{\beta}\,(r) - i\,y\overline{\beta}\,(r) + \delta\,(r)$$

$\Phi(t, r) = \Phi$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{2}\sqrt{2}\Phi \right) dt + \left(-\frac{\sqrt{2}\Phi \frac{\partial}{\partial r} P}{2\,P} + \frac{1}{2}\sqrt{2} \frac{\partial}{\partial r} \Phi \right) dr$$

$$n = \left(\frac{1}{2}\sqrt{2}\Phi \right) dt + \left(\frac{\sqrt{2}\Phi \frac{\partial}{\partial r} P}{2\,P} - \frac{1}{2}\sqrt{2} \frac{\partial}{\partial r} \Phi \right) dr$$

$$m = \left(\frac{\sqrt{2}\Phi}{2\,P} \right) dx + \left(\frac{i\sqrt{2}\Phi}{2\,P} \right) dy$$

$$\overline{m} = \left(\frac{\sqrt{2}\Phi}{2\,P}\right)\,dx + \left(-\frac{i\,\sqrt{2}\Phi}{2\,P}\right)\,dy$$

Contravariant components

$$l = \left(-\frac{\sqrt{2}}{2\,\Phi}\right)\,dt + \left(-\frac{P}{\sqrt{2}\Phi\frac{\partial}{\partial r}P - \sqrt{2}P\frac{\partial}{\partial r}\Phi}\right)\,dr$$

$$n = \left(-\frac{\sqrt{2}}{2\,\Phi}\right)\,dt + \left(\frac{P}{\sqrt{2}\Phi\frac{\partial}{\partial r}P - \sqrt{2}P\frac{\partial}{\partial r}\Phi}\right)\,dr$$

$$m = \left(\frac{\sqrt{2}P}{2\,\Phi}\right)\,dx + \left(\frac{i\,\sqrt{2}P}{2\,\Phi}\right)\,dy$$

$$\overline{m} = \left(\frac{\sqrt{2}P}{2\,\Phi}\right)\,dx + \left(-\frac{i\,\sqrt{2}P}{2\,\Phi}\right)\,dy$$

Directional Derivatives

$$D = -\frac{\sqrt{2}}{2\,\Phi}\,\partial_t - \frac{P}{\sqrt{2}\Phi\frac{\partial}{\partial r}P - \sqrt{2}P\frac{\partial}{\partial r}\Phi}\,\partial_r$$

$$\Delta = -\frac{\sqrt{2}}{2\,\Phi}\,\partial_t + \frac{P}{\sqrt{2}\Phi\frac{\partial}{\partial r}P - \sqrt{2}P\frac{\partial}{\partial r}\Phi}\,\partial_r$$

$$\delta = \frac{\sqrt{2}P}{2\,\Phi}\,\partial_x + \frac{i\,\sqrt{2}P}{2\,\Phi}\,\partial_y$$

$$\overline{\delta} = \frac{\sqrt{2}P}{2\,\Phi}\,\partial_x - \frac{i\,\sqrt{2}P}{2\,\Phi}\,\partial_y$$

Spin Coefficients

$$\rho = -\frac{\sqrt{2}}{2\,\Phi} + \frac{\sqrt{2}\frac{\partial}{\partial t}\Phi}{2\,\Phi^2}$$

$$\sigma = 0$$

$$\kappa = \frac{\sqrt{2}P\frac{\partial^2}{\partial x\partial r}P}{4\left(\Phi\frac{\partial}{\partial r}P - P\frac{\partial}{\partial r}\Phi\right)} + \frac{i\,\sqrt{2}P\frac{\partial^2}{\partial y\partial r}P}{4\left(\Phi\frac{\partial}{\partial r}P - P\frac{\partial}{\partial r}\Phi\right)} - \frac{\sqrt{2}\frac{\partial}{\partial x}P\frac{\partial}{\partial r}P}{4\left(\Phi\frac{\partial}{\partial r}P - P\frac{\partial}{\partial r}\Phi\right)} - \frac{i\,\sqrt{2}\frac{\partial}{\partial y}P\frac{\partial}{\partial r}P}{4\left(\Phi\frac{\partial}{\partial r}P - P\frac{\partial}{\partial r}\Phi\right)}$$

$$\mu = -\frac{\sqrt{2}}{2\,\Phi} - \frac{\sqrt{2}\frac{\partial}{\partial t}\Phi}{2\,\Phi^2}$$

$$\lambda = 0$$

$$\nu = -\frac{\sqrt{2}P\frac{\partial^2}{\partial x\partial r}P}{4\left(\Phi\frac{\partial}{\partial r}P - P\frac{\partial}{\partial r}\Phi\right)} + \frac{i\,\sqrt{2}P\frac{\partial^2}{\partial y\partial r}P}{4\left(\Phi\frac{\partial}{\partial r}P - P\frac{\partial}{\partial r}\Phi\right)} + \frac{\sqrt{2}\frac{\partial}{\partial x}P\frac{\partial}{\partial r}P}{4\left(\Phi\frac{\partial}{\partial r}P - P\frac{\partial}{\partial r}\Phi\right)} - \frac{i\,\sqrt{2}\frac{\partial}{\partial y}P\frac{\partial}{\partial r}P}{4\left(\Phi\frac{\partial}{\partial r}P - P\frac{\partial}{\partial r}\Phi\right)}$$

$$\alpha = \frac{\sqrt{2}\frac{\partial}{\partial x}P}{4\,\Phi} - \frac{i\,\sqrt{2}\frac{\partial}{\partial y}P}{4\,\Phi}$$

$$\beta = -\frac{\sqrt{2}\frac{\partial}{\partial x}P}{4\,\Phi} - \frac{i\,\sqrt{2}\frac{\partial}{\partial y}P}{4\,\Phi}$$

$$\tau = -\frac{\sqrt{2}P\frac{\partial^2}{\partial x\partial r}P}{4\left(\Phi\frac{\partial}{\partial r}P - P\frac{\partial}{\partial r}\Phi\right)} - \frac{i\,\sqrt{2}P\frac{\partial^2}{\partial y\partial r}P}{4\left(\Phi\frac{\partial}{\partial r}P - P\frac{\partial}{\partial r}\Phi\right)} + \frac{\sqrt{2}\frac{\partial}{\partial x}P\frac{\partial}{\partial r}P}{4\left(\Phi\frac{\partial}{\partial r}P - P\frac{\partial}{\partial r}\Phi\right)} + \frac{i\,\sqrt{2}\frac{\partial}{\partial y}P\frac{\partial}{\partial r}P}{4\left(\Phi\frac{\partial}{\partial r}P - P\frac{\partial}{\partial r}\Phi\right)}$$

$$\pi = \frac{\sqrt{2}P\frac{\partial^2}{\partial x\partial r}P}{4\left(\Phi\frac{\partial}{\partial r}P - P\frac{\partial}{\partial r}\Phi\right)} - \frac{i\,\sqrt{2}P\frac{\partial^2}{\partial y\partial r}P}{4\left(\Phi\frac{\partial}{\partial r}P - P\frac{\partial}{\partial r}\Phi\right)} - \frac{\sqrt{2}\frac{\partial}{\partial x}P\frac{\partial}{\partial r}P}{4\left(\Phi\frac{\partial}{\partial r}P - P\frac{\partial}{\partial r}\Phi\right)} + \frac{i\,\sqrt{2}\frac{\partial}{\partial y}P\frac{\partial}{\partial r}P}{4\left(\Phi\frac{\partial}{\partial r}P - P\frac{\partial}{\partial r}\Phi\right)}$$

$$\epsilon = -\frac{\frac{\partial}{\partial r}P\frac{\partial}{\partial t}\Phi}{2\left(\sqrt{2}\Phi^2\frac{\partial}{\partial r}P-\sqrt{2}P\Phi\frac{\partial}{\partial r}\Phi\right)}+\frac{P\frac{\partial^2}{\partial t\partial r}\Phi}{2\left(\sqrt{2}\Phi^2\frac{\partial}{\partial r}P-\sqrt{2}P\Phi\frac{\partial}{\partial r}\Phi\right)}-\frac{P\frac{\partial}{\partial r}\Phi}{2\left(\sqrt{2}\Phi^2\frac{\partial}{\partial r}P-\sqrt{2}P\Phi\frac{\partial}{\partial r}\Phi\right)}$$

$$\gamma = \frac{\frac{\partial}{\partial r}P\frac{\partial}{\partial t}\Phi}{2\left(\sqrt{2}\Phi^2\frac{\partial}{\partial r}P-\sqrt{2}P\Phi\frac{\partial}{\partial r}\Phi\right)}-\frac{P\frac{\partial^2}{\partial t\partial r}\Phi}{2\left(\sqrt{2}\Phi^2\frac{\partial}{\partial r}P-\sqrt{2}P\Phi\frac{\partial}{\partial r}\Phi\right)}-\frac{P\frac{\partial}{\partial r}\Phi}{2\left(\sqrt{2}\Phi^2\frac{\partial}{\partial r}P-\sqrt{2}P\Phi\frac{\partial}{\partial r}\Phi\right)}$$

Ricci Tensor Components

$$\begin{aligned}\Phi_{00} = &-\frac{P^2\Phi^3\frac{\partial^3}{(\partial x)^2\partial r}P}{4\left(\Phi^5\frac{\partial}{\partial r}P-P\Phi^4\frac{\partial}{\partial r}\Phi\right)}+\frac{P\Phi^3\frac{\partial}{\partial x}P\frac{\partial^2}{\partial x\partial r}P}{2\left(\Phi^5\frac{\partial}{\partial r}P-P\Phi^4\frac{\partial}{\partial r}\Phi\right)}-\frac{P^2\Phi^3\frac{\partial^3}{(\partial y)^2\partial r}P}{4\left(\Phi^5\frac{\partial}{\partial r}P-P\Phi^4\frac{\partial}{\partial r}\Phi\right)}+\frac{P\Phi^3\frac{\partial}{\partial y}P\frac{\partial^2}{\partial y\partial r}P}{2\left(\Phi^5\frac{\partial}{\partial r}P-P\Phi^4\frac{\partial}{\partial r}\Phi\right)}-\frac{\Phi^3\frac{\partial}{\partial x}P^2\frac{\partial}{\partial r}P}{2\left(\Phi^5\frac{\partial}{\partial r}P-P\Phi^4\frac{\partial}{\partial r}\Phi\right)}\\
&+\frac{P\Phi^3\frac{\partial^2}{(\partial x)^2}P\frac{\partial}{\partial r}P}{4\left(\Phi^5\frac{\partial}{\partial r}P-P\Phi^4\frac{\partial}{\partial r}\Phi\right)}-\frac{\Phi^3\frac{\partial}{\partial y}P^2\frac{\partial}{\partial r}P}{2\left(\Phi^5\frac{\partial}{\partial r}P-P\Phi^4\frac{\partial}{\partial r}\Phi\right)}+\frac{P\Phi^3\frac{\partial^2}{(\partial y)^2}P\frac{\partial}{\partial r}P}{4\left(\Phi^5\frac{\partial}{\partial r}P-P\Phi^4\frac{\partial}{\partial r}\Phi\right)}-\frac{\Phi^3\frac{\partial}{\partial r}P}{2\left(\Phi^5\frac{\partial}{\partial r}P-P\Phi^4\frac{\partial}{\partial r}\Phi\right)}+\frac{\Phi\frac{\partial}{\partial r}P\frac{\partial}{\partial t}\Phi^2}{\Phi^5\frac{\partial}{\partial r}P-P\Phi^4\frac{\partial}{\partial r}\Phi}\\
&-\frac{\Phi^2\frac{\partial}{\partial r}P\frac{\partial^2}{(\partial t)^2}\Phi}{2\left(\Phi^5\frac{\partial}{\partial r}P-P\Phi^4\frac{\partial}{\partial r}\Phi\right)}-\frac{P\Phi\frac{\partial}{\partial t}\Phi\frac{\partial^2}{\partial t\partial r}\Phi}{2\left(\Phi^5\frac{\partial}{\partial r}P-P\Phi^4\frac{\partial}{\partial r}\Phi\right)}-\frac{P\Phi^2\frac{\partial}{\partial r}\Phi}{2\left(\Phi^5\frac{\partial}{\partial r}P-P\Phi^4\frac{\partial}{\partial r}\Phi\right)}+\frac{P\Phi\frac{\partial}{\partial t}\Phi\frac{\partial}{\partial r}\Phi}{\Phi^5\frac{\partial}{\partial r}P-P\Phi^4\frac{\partial}{\partial r}\Phi}-\frac{P\frac{\partial}{\partial t}\Phi^2\frac{\partial}{\partial r}\Phi}{2\left(\Phi^5\frac{\partial}{\partial r}P-P\Phi^4\frac{\partial}{\partial r}\Phi\right)}+\frac{P\Phi\frac{\partial^2}{(\partial t)^2}\Phi\frac{\partial}{\partial r}\Phi}{2\left(\Phi^5\frac{\partial}{\partial r}P-P\Phi^4\frac{\partial}{\partial r}\Phi\right)}\end{aligned}$$

$$\begin{aligned}\Phi_{01} = &-\frac{P^2\frac{\partial^2}{\partial x\partial r}P\frac{\partial}{\partial r}\Phi}{4\left(\Phi^3\frac{\partial}{\partial r}P^2-2\,P\Phi^2\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi+P^2\Phi\frac{\partial}{\partial r}\Phi^2\right)}-\frac{i\,P^2\frac{\partial^2}{\partial y\partial r}P\frac{\partial}{\partial r}\Phi}{4\left(\Phi^3\frac{\partial}{\partial r}P^2-2\,P\Phi^2\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi+P^2\Phi\frac{\partial}{\partial r}\Phi^2\right)}\\
&+\frac{P\frac{\partial}{\partial x}P\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi}{4\left(\Phi^3\frac{\partial}{\partial r}P^2-2\,P\Phi^2\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi+P^2\Phi\frac{\partial}{\partial r}\Phi^2\right)}+\frac{i\,P\frac{\partial}{\partial y}P\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi}{4\left(\Phi^3\frac{\partial}{\partial r}P^2-2\,P\Phi^2\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi+P^2\Phi\frac{\partial}{\partial r}\Phi^2\right)}\end{aligned}$$

$$\Phi_{02} = -\frac{P^2\frac{\partial^3}{(\partial x)^2\partial r}P}{4\left(\Phi^2\frac{\partial}{\partial r}P-P\Phi\frac{\partial}{\partial r}\Phi\right)}-\frac{i\,P^2\frac{\partial^3}{\partial x\partial y\partial r}P}{2\left(\Phi^2\frac{\partial}{\partial r}P-P\Phi\frac{\partial}{\partial r}\Phi\right)}+\frac{P^2\frac{\partial^3}{(\partial y)^2\partial r}P}{4\left(\Phi^2\frac{\partial}{\partial r}P-P\Phi\frac{\partial}{\partial r}\Phi\right)}+\frac{P\frac{\partial^2}{(\partial x)^2}P\frac{\partial}{\partial r}P}{4\left(\Phi^2\frac{\partial}{\partial r}P-P\Phi\frac{\partial}{\partial r}\Phi\right)}+\frac{i\,P\frac{\partial^2}{\partial x\partial y}P\frac{\partial}{\partial r}P}{2\left(\Phi^2\frac{\partial}{\partial r}P-P\Phi\frac{\partial}{\partial r}\Phi\right)}-\frac{P\frac{\partial^2}{(\partial y)^2}P\frac{\partial}{\partial r}P}{4\left(\Phi^2\frac{\partial}{\partial r}P-P\Phi\frac{\partial}{\partial r}\Phi\right)}$$

$$\begin{aligned}\Phi_{10} = &-\frac{P^2\frac{\partial^2}{\partial x\partial r}P\frac{\partial}{\partial r}\Phi}{4\left(\Phi^3\frac{\partial}{\partial r}P^2-2\,P\Phi^2\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi+P^2\Phi\frac{\partial}{\partial r}\Phi^2\right)}+\frac{i\,P^2\frac{\partial^2}{\partial y\partial r}P\frac{\partial}{\partial r}\Phi}{4\left(\Phi^3\frac{\partial}{\partial r}P^2-2\,P\Phi^2\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi+P^2\Phi\frac{\partial}{\partial r}\Phi^2\right)}\\
&+\frac{P\frac{\partial}{\partial x}P\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi}{4\left(\Phi^3\frac{\partial}{\partial r}P^2-2\,P\Phi^2\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi+P^2\Phi\frac{\partial}{\partial r}\Phi^2\right)}-\frac{i\,P\frac{\partial}{\partial y}P\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi}{4\left(\Phi^3\frac{\partial}{\partial r}P^2-2\,P\Phi^2\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi+P^2\Phi\frac{\partial}{\partial r}\Phi^2\right)}\end{aligned}$$

$$
\begin{aligned}
\Phi_{11} = & -\frac{\Phi^5 \frac{\partial}{\partial x} P^2 \frac{\partial}{\partial r} P^3}{4\left(\Phi^7 \frac{\partial}{\partial r} P^3 - 3\,P\Phi^6 \frac{\partial}{\partial r} P^2 \frac{\partial}{\partial r}\Phi + 3\,P^2\Phi^5 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi^2 - P^3\Phi^4 \frac{\partial}{\partial r}\Phi^3\right)} + \frac{P\Phi^5 \frac{\partial^2}{(\partial x)^2} P \frac{\partial}{\partial r} P^3}{4\left(\Phi^7 \frac{\partial}{\partial r} P^3 - 3\,P\Phi^6 \frac{\partial}{\partial r} P^2 \frac{\partial}{\partial r}\Phi + 3\,P^2\Phi^5 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi^2 - P^3\Phi^4 \frac{\partial}{\partial r}\Phi^3\right)} - \frac{\Phi^5 \frac{\partial}{\partial y} P^2 \frac{\partial}{\partial r} P^3}{4\left(\Phi^7 \frac{\partial}{\partial r} P^3 - 3\,P\Phi^6 \frac{\partial}{\partial r} P^2 \frac{\partial}{\partial r}\Phi + 3\,P^2\Phi^5 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi^2 - P^3\Phi^4 \frac{\partial}{\partial r}\Phi^3\right)} \\
& + \frac{P\Phi^5 \frac{\partial^2}{(\partial y)^2} P \frac{\partial}{\partial r} P^3}{4\left(\Phi^7 \frac{\partial}{\partial r} P^3 - 3\,P\Phi^6 \frac{\partial}{\partial r} P^2 \frac{\partial}{\partial r}\Phi + 3\,P^2\Phi^5 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi^2 - P^3\Phi^4 \frac{\partial}{\partial r}\Phi^3\right)} + \frac{3\,P\Phi^4 \frac{\partial}{\partial x} P^2 \frac{\partial}{\partial r} P^2 \frac{\partial}{\partial r}\Phi}{4\left(\Phi^7 \frac{\partial}{\partial r} P^3 - 3\,P\Phi^6 \frac{\partial}{\partial r} P^2 \frac{\partial}{\partial r}\Phi + 3\,P^2\Phi^5 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi^2 - P^3\Phi^4 \frac{\partial}{\partial r}\Phi^3\right)} \\
& - \frac{3\,P^2\Phi^4 \frac{\partial^2}{(\partial x)^2} P \frac{\partial}{\partial r} P^2 \frac{\partial}{\partial r}\Phi}{4\left(\Phi^7 \frac{\partial}{\partial r} P^3 - 3\,P\Phi^6 \frac{\partial}{\partial r} P^2 \frac{\partial}{\partial r}\Phi + 3\,P^2\Phi^5 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi^2 - P^3\Phi^4 \frac{\partial}{\partial r}\Phi^3\right)} + \frac{3\,P\Phi^4 \frac{\partial}{\partial y} P^2 \frac{\partial}{\partial r} P^2 \frac{\partial}{\partial r}\Phi}{4\left(\Phi^7 \frac{\partial}{\partial r} P^3 - 3\,P\Phi^6 \frac{\partial}{\partial r} P^2 \frac{\partial}{\partial r}\Phi + 3\,P^2\Phi^5 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi^2 - P^3\Phi^4 \frac{\partial}{\partial r}\Phi^3\right)} - \frac{3\,P^2\Phi^4 \frac{\partial^2}{(\partial y)^2} P \frac{\partial}{\partial r} P^2 \frac{\partial}{\partial r}\Phi}{4\left(\Phi^7 \frac{\partial}{\partial r} P^3 - 3\,P\Phi^6 \frac{\partial}{\partial r} P^2 \frac{\partial}{\partial r}\Phi + 3\,P^2\Phi^5 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi^2 - P^3\Phi^4 \frac{\partial}{\partial r}\Phi^3\right)} \\
& - \frac{3\,P^2\Phi^3 \frac{\partial}{\partial x} P^2 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi^2}{4\left(\Phi^7 \frac{\partial}{\partial r} P^3 - 3\,P\Phi^6 \frac{\partial}{\partial r} P^2 \frac{\partial}{\partial r}\Phi + 3\,P^2\Phi^5 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi^2 - P^3\Phi^4 \frac{\partial}{\partial r}\Phi^3\right)} + \frac{3\,P^3\Phi^3 \frac{\partial^2}{(\partial x)^2} P \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi^2}{4\left(\Phi^7 \frac{\partial}{\partial r} P^3 - 3\,P\Phi^6 \frac{\partial}{\partial r} P^2 \frac{\partial}{\partial r}\Phi + 3\,P^2\Phi^5 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi^2 - P^3\Phi^4 \frac{\partial}{\partial r}\Phi^3\right)} - \frac{3\,P^2\Phi^3 \frac{\partial}{\partial y} P^2 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi^2}{4\left(\Phi^7 \frac{\partial}{\partial r} P^3 - 3\,P\Phi^6 \frac{\partial}{\partial r} P^2 \frac{\partial}{\partial r}\Phi + 3\,P^2\Phi^5 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi^2 - P^3\Phi^4 \frac{\partial}{\partial r}\Phi^3\right)} \\
& + \frac{3\,P^3\Phi^3 \frac{\partial^2}{(\partial y)^2} P \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi^2}{4\left(\Phi^7 \frac{\partial}{\partial r} P^3 - 3\,P\Phi^6 \frac{\partial}{\partial r} P^2 \frac{\partial}{\partial r}\Phi + 3\,P^2\Phi^5 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi^2 - P^3\Phi^4 \frac{\partial}{\partial r}\Phi^3\right)} + \frac{P^3\Phi^2 \frac{\partial}{\partial x} P^2 \frac{\partial}{\partial r}\Phi^3}{4\left(\Phi^7 \frac{\partial}{\partial r} P^3 - 3\,P\Phi^6 \frac{\partial}{\partial r} P^2 \frac{\partial}{\partial r}\Phi + 3\,P^2\Phi^5 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi^2 - P^3\Phi^4 \frac{\partial}{\partial r}\Phi^3\right)} - \frac{P^4\Phi^2 \frac{\partial^2}{(\partial x)^2} P \frac{\partial}{\partial r}\Phi^3}{4\left(\Phi^7 \frac{\partial}{\partial r} P^3 - 3\,P\Phi^6 \frac{\partial}{\partial r} P^2 \frac{\partial}{\partial r}\Phi + 3\,P^2\Phi^5 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi^2 - P^3\Phi^4 \frac{\partial}{\partial r}\Phi^3\right)} \\
& + \frac{P^3\Phi^2 \frac{\partial}{\partial y} P^2 \frac{\partial}{\partial r}\Phi^3}{4\left(\Phi^7 \frac{\partial}{\partial r} P^3 - 3\,P\Phi^6 \frac{\partial}{\partial r} P^2 \frac{\partial}{\partial r}\Phi + 3\,P^2\Phi^5 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi^2 - P^3\Phi^4 \frac{\partial}{\partial r}\Phi^3\right)} - \frac{P^4\Phi^2 \frac{\partial^2}{(\partial y)^2} P \frac{\partial}{\partial r}\Phi^3}{4\left(\Phi^7 \frac{\partial}{\partial r} P^3 - 3\,P\Phi^6 \frac{\partial}{\partial r} P^2 \frac{\partial}{\partial r}\Phi + 3\,P^2\Phi^5 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi^2 - P^3\Phi^4 \frac{\partial}{\partial r}\Phi^3\right)} - \frac{\Phi^5 \frac{\partial}{\partial r} P^3}{4\left(\Phi^7 \frac{\partial}{\partial r} P^3 - 3\,P\Phi^6 \frac{\partial}{\partial r} P^2 \frac{\partial}{\partial r}\Phi + 3\,P^2\Phi^5 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi^2 - P^3\Phi^4 \frac{\partial}{\partial r}\Phi^3\right)} \\
& + \frac{\Phi^3 \frac{\partial}{\partial r} P^3 \frac{\partial}{\partial t}\Phi^2}{2\left(\Phi^7 \frac{\partial}{\partial r} P^3 - 3\,P\Phi^6 \frac{\partial}{\partial r} P^2 \frac{\partial}{\partial r}\Phi + 3\,P^2\Phi^5 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi^2 - P^3\Phi^4 \frac{\partial}{\partial r}\Phi^3\right)} - \frac{\Phi^4 \frac{\partial}{\partial r} P^3 \frac{\partial^2}{(\partial t)^2}\Phi}{4\left(\Phi^7 \frac{\partial}{\partial r} P^3 - 3\,P\Phi^6 \frac{\partial}{\partial r} P^2 \frac{\partial}{\partial r}\Phi + 3\,P^2\Phi^5 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi^2 - P^3\Phi^4 \frac{\partial}{\partial r}\Phi^3\right)} + \frac{P\Phi^4 \frac{\partial}{\partial r} P^2 \frac{\partial^3}{(\partial t)^2\partial r}\Phi}{4\left(\Phi^7 \frac{\partial}{\partial r} P^3 - 3\,P\Phi^6 \frac{\partial}{\partial r} P^2 \frac{\partial}{\partial r}\Phi + 3\,P^2\Phi^5 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi^2 - P^3\Phi^4 \frac{\partial}{\partial r}\Phi^3\right)} \\
& - \frac{P\Phi^3 \frac{\partial}{\partial r} P^2 \frac{\partial}{\partial t}\Phi \frac{\partial^2}{\partial t\partial r}\Phi}{4\left(\Phi^7 \frac{\partial}{\partial r} P^3 - 3\,P\Phi^6 \frac{\partial}{\partial r} P^2 \frac{\partial}{\partial r}\Phi + 3\,P^2\Phi^5 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi^2 - P^3\Phi^4 \frac{\partial}{\partial r}\Phi^3\right)} + \frac{P\Phi^4 \frac{\partial}{\partial r} P^2 \frac{\partial}{\partial r}\Phi}{\Phi^7 \frac{\partial}{\partial r} P^3 - 3\,P\Phi^6 \frac{\partial}{\partial r} P^2 \frac{\partial}{\partial r}\Phi + 3\,P^2\Phi^5 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi^2 - P^3\Phi^4 \frac{\partial}{\partial r}\Phi^3} - \frac{P^2\Phi^4 \frac{\partial^2}{(\partial r)^2} P \frac{\partial}{\partial r}\Phi}{4\left(\Phi^7 \frac{\partial}{\partial r} P^3 - 3\,P\Phi^6 \frac{\partial}{\partial r} P^2 \frac{\partial}{\partial r}\Phi + 3\,P^2\Phi^5 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi^2 - P^3\Phi^4 \frac{\partial}{\partial r}\Phi^3\right)} \\
& - \frac{5\,P\Phi^2 \frac{\partial}{\partial r} P^2 \frac{\partial}{\partial t}\Phi^2 \frac{\partial}{\partial r}\Phi}{4\left(\Phi^7 \frac{\partial}{\partial r} P^3 - 3\,P\Phi^6 \frac{\partial}{\partial r} P^2 \frac{\partial}{\partial r}\Phi + 3\,P^2\Phi^5 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi^2 - P^3\Phi^4 \frac{\partial}{\partial r}\Phi^3\right)} + \frac{P\Phi^3 \frac{\partial}{\partial r} P^2 \frac{\partial^2}{(\partial t)^2}\Phi \frac{\partial}{\partial r}\Phi}{2\left(\Phi^7 \frac{\partial}{\partial r} P^3 - 3\,P\Phi^6 \frac{\partial}{\partial r} P^2 \frac{\partial}{\partial r}\Phi + 3\,P^2\Phi^5 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi^2 - P^3\Phi^4 \frac{\partial}{\partial r}\Phi^3\right)} - \frac{P^2\Phi^3 \frac{\partial}{\partial r} P \frac{\partial^3}{(\partial t)^2\partial r}\Phi \frac{\partial}{\partial r}\Phi}{2\left(\Phi^7 \frac{\partial}{\partial r} P^3 - 3\,P\Phi^6 \frac{\partial}{\partial r} P^2 \frac{\partial}{\partial r}\Phi + 3\,P^2\Phi^5 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi^2 - P^3\Phi^4 \frac{\partial}{\partial r}\Phi^3\right)} \\
& + \frac{P^2\Phi^2 \frac{\partial}{\partial r} P \frac{\partial}{\partial t}\Phi \frac{\partial^2}{\partial t\partial r}\Phi \frac{\partial}{\partial r}\Phi}{2\left(\Phi^7 \frac{\partial}{\partial r} P^3 - 3\,P\Phi^6 \frac{\partial}{\partial r} P^2 \frac{\partial}{\partial r}\Phi + 3\,P^2\Phi^5 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi^2 - P^3\Phi^4 \frac{\partial}{\partial r}\Phi^3\right)} - \frac{P^2\Phi^3 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi^2}{\Phi^7 \frac{\partial}{\partial r} P^3 - 3\,P\Phi^6 \frac{\partial}{\partial r} P^2 \frac{\partial}{\partial r}\Phi + 3\,P^2\Phi^5 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi^2 - P^3\Phi^4 \frac{\partial}{\partial r}\Phi^3} + \frac{P^2\Phi \frac{\partial}{\partial r} P \frac{\partial}{\partial t}\Phi^2 \frac{\partial}{\partial r}\Phi^2}{\Phi^7 \frac{\partial}{\partial r} P^3 - 3\,P\Phi^6 \frac{\partial}{\partial r} P^2 \frac{\partial}{\partial r}\Phi + 3\,P^2\Phi^5 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi^2 - P^3\Phi^4 \frac{\partial}{\partial r}\Phi^3} \\
& - \frac{P^2\Phi^2 \frac{\partial}{\partial r} P \frac{\partial^2}{(\partial t)^2}\Phi \frac{\partial}{\partial r}\Phi^2}{4\left(\Phi^7 \frac{\partial}{\partial r} P^3 - 3\,P\Phi^6 \frac{\partial}{\partial r} P^2 \frac{\partial}{\partial r}\Phi + 3\,P^2\Phi^5 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi^2 - P^3\Phi^4 \frac{\partial}{\partial r}\Phi^3\right)} + \frac{P^3\Phi^2 \frac{\partial^3}{(\partial t)^2\partial r}\Phi \frac{\partial}{\partial r}\Phi^2}{4\left(\Phi^7 \frac{\partial}{\partial r} P^3 - 3\,P\Phi^6 \frac{\partial}{\partial r} P^2 \frac{\partial}{\partial r}\Phi + 3\,P^2\Phi^5 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi^2 - P^3\Phi^4 \frac{\partial}{\partial r}\Phi^3\right)} - \frac{P^3\Phi \frac{\partial}{\partial t}\Phi \frac{\partial^2}{\partial t\partial r}\Phi \frac{\partial}{\partial r}\Phi^2}{4\left(\Phi^7 \frac{\partial}{\partial r} P^3 - 3\,P\Phi^6 \frac{\partial}{\partial r} P^2 \frac{\partial}{\partial r}\Phi + 3\,P^2\Phi^5 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi^2 - P^3\Phi^4 \frac{\partial}{\partial r}\Phi^3\right)} \\
& + \frac{P^3\Phi^2 \frac{\partial}{\partial r}\Phi^3}{4\left(\Phi^7 \frac{\partial}{\partial r} P^3 - 3\,P\Phi^6 \frac{\partial}{\partial r} P^2 \frac{\partial}{\partial r}\Phi + 3\,P^2\Phi^5 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi^2 - P^3\Phi^4 \frac{\partial}{\partial r}\Phi^3\right)} - \frac{P^3 \frac{\partial}{\partial t}\Phi^2 \frac{\partial}{\partial r}\Phi^3}{4\left(\Phi^7 \frac{\partial}{\partial r} P^3 - 3\,P\Phi^6 \frac{\partial}{\partial r} P^2 \frac{\partial}{\partial r}\Phi + 3\,P^2\Phi^5 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi^2 - P^3\Phi^4 \frac{\partial}{\partial r}\Phi^3\right)} + \frac{P^2\Phi^4 \frac{\partial}{\partial r} P \frac{\partial^2}{(\partial r)^2}\Phi}{4\left(\Phi^7 \frac{\partial}{\partial r} P^3 - 3\,P\Phi^6 \frac{\partial}{\partial r} P^2 \frac{\partial}{\partial r}\Phi + 3\,P^2\Phi^5 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi^2 - P^3\Phi^4 \frac{\partial}{\partial r}\Phi^3\right)}
\end{aligned}
$$

$$
\begin{aligned}
\Phi_{12} = & \frac{P^2 \frac{\partial^2}{\partial x\partial r} P \frac{\partial}{\partial r}\Phi}{4\left(\Phi^3 \frac{\partial}{\partial r} P^2 - 2\,P\Phi^2 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi + P^2\Phi \frac{\partial}{\partial r}\Phi^2\right)} + \frac{i\,P^2 \frac{\partial^2}{\partial y\partial r} P \frac{\partial}{\partial r}\Phi}{4\left(\Phi^3 \frac{\partial}{\partial r} P^2 - 2\,P\Phi^2 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi + P^2\Phi \frac{\partial}{\partial r}\Phi^2\right)} \\
& - \frac{P \frac{\partial}{\partial x} P \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi}{4\left(\Phi^3 \frac{\partial}{\partial r} P^2 - 2\,P\Phi^2 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi + P^2\Phi \frac{\partial}{\partial r}\Phi^2\right)} - \frac{i\,P \frac{\partial}{\partial y} P \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi}{4\left(\Phi^3 \frac{\partial}{\partial r} P^2 - 2\,P\Phi^2 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi + P^2\Phi \frac{\partial}{\partial r}\Phi^2\right)}
\end{aligned}
$$

$$
\Phi_{20} = -\frac{P^2 \frac{\partial^3}{(\partial x)^2\partial r} P}{4\left(\Phi^2 \frac{\partial}{\partial r} P - P\Phi \frac{\partial}{\partial r}\Phi\right)} + \frac{i\,P^2 \frac{\partial^3}{\partial x\partial y\partial r} P}{2\left(\Phi^2 \frac{\partial}{\partial r} P - P\Phi \frac{\partial}{\partial r}\Phi\right)} + \frac{P^2 \frac{\partial^3}{(\partial y)^2\partial r} P}{4\left(\Phi^2 \frac{\partial}{\partial r} P - P\Phi \frac{\partial}{\partial r}\Phi\right)} + \frac{P \frac{\partial^2}{(\partial x)^2} P \frac{\partial}{\partial r} P}{4\left(\Phi^2 \frac{\partial}{\partial r} P - P\Phi \frac{\partial}{\partial r}\Phi\right)} - \frac{i\,P \frac{\partial^2}{\partial x\partial y} P \frac{\partial}{\partial r} P}{2\left(\Phi^2 \frac{\partial}{\partial r} P - P\Phi \frac{\partial}{\partial r}\Phi\right)} - \frac{P \frac{\partial^2}{(\partial y)^2} P \frac{\partial}{\partial r} P}{4\left(\Phi^2 \frac{\partial}{\partial r} P - P\Phi \frac{\partial}{\partial r}\Phi\right)}
$$

$$\Phi_{21} = \frac{P^2 \frac{\partial^2}{\partial x \partial r} P \frac{\partial}{\partial r} \Phi}{4 \left(\Phi^3 \frac{\partial}{\partial r} P^2 - 2\, P \Phi^2 \frac{\partial}{\partial r} P \frac{\partial}{\partial r} \Phi + P^2 \Phi \frac{\partial}{\partial r} \Phi^2\right)} - \frac{i\, P^2 \frac{\partial^2}{\partial y \partial r} P \frac{\partial}{\partial r} \Phi}{4 \left(\Phi^3 \frac{\partial}{\partial r} P^2 - 2\, P \Phi^2 \frac{\partial}{\partial r} P \frac{\partial}{\partial r} \Phi + P^2 \Phi \frac{\partial}{\partial r} \Phi^2\right)}$$
$$- \frac{P \frac{\partial}{\partial x} P \frac{\partial}{\partial r} P \frac{\partial}{\partial r} \Phi}{4 \left(\Phi^3 \frac{\partial}{\partial r} P^2 - 2\, P \Phi^2 \frac{\partial}{\partial r} P \frac{\partial}{\partial r} \Phi + P^2 \Phi \frac{\partial}{\partial r} \Phi^2\right)} + \frac{i\, P \frac{\partial}{\partial y} P \frac{\partial}{\partial r} P \frac{\partial}{\partial r} \Phi}{4 \left(\Phi^3 \frac{\partial}{\partial r} P^2 - 2\, P \Phi^2 \frac{\partial}{\partial r} P \frac{\partial}{\partial r} \Phi + P^2 \Phi \frac{\partial}{\partial r} \Phi^2\right)}$$

$$\Phi_{22} = - \frac{P^2 \Phi^3 \frac{\partial^3}{(\partial x)^2 \partial r} P}{4 \left(\Phi^5 \frac{\partial}{\partial r} P - P \Phi^4 \frac{\partial}{\partial r} \Phi\right)} + \frac{P \Phi^3 \frac{\partial}{\partial x} P \frac{\partial^2}{\partial x \partial r} P}{2 \left(\Phi^5 \frac{\partial}{\partial r} P - P \Phi^4 \frac{\partial}{\partial r} \Phi\right)} - \frac{P^2 \Phi^3 \frac{\partial^3}{(\partial y)^2 \partial r} P}{4 \left(\Phi^5 \frac{\partial}{\partial r} P - P \Phi^4 \frac{\partial}{\partial r} \Phi\right)} + \frac{P \Phi^3 \frac{\partial}{\partial y} P \frac{\partial^2}{\partial y \partial r} P}{2 \left(\Phi^5 \frac{\partial}{\partial r} P - P \Phi^4 \frac{\partial}{\partial r} \Phi\right)} - \frac{\Phi^3 \frac{\partial}{\partial x} P^2 \frac{\partial}{\partial r} P}{2 \left(\Phi^5 \frac{\partial}{\partial r} P - P \Phi^4 \frac{\partial}{\partial r} \Phi\right)}$$
$$+ \frac{P \Phi^3 \frac{\partial^2}{(\partial x)^2} P \frac{\partial}{\partial r} P}{4 \left(\Phi^5 \frac{\partial}{\partial r} P - P \Phi^4 \frac{\partial}{\partial r} \Phi\right)} - \frac{\Phi^3 \frac{\partial}{\partial y} P^2 \frac{\partial}{\partial r} P}{2 \left(\Phi^5 \frac{\partial}{\partial r} P - P \Phi^4 \frac{\partial}{\partial r} \Phi\right)} + \frac{P \Phi^3 \frac{\partial^2}{(\partial y)^2} P \frac{\partial}{\partial r} P}{4 \left(\Phi^5 \frac{\partial}{\partial r} P - P \Phi^4 \frac{\partial}{\partial r} \Phi\right)} - \frac{\Phi^3 \frac{\partial}{\partial r} P}{2 \left(\Phi^5 \frac{\partial}{\partial r} P - P \Phi^4 \frac{\partial}{\partial r} \Phi\right)} + \frac{\Phi \frac{\partial}{\partial r} P \frac{\partial}{\partial t} \Phi^2}{\Phi^5 \frac{\partial}{\partial r} P - P \Phi^4 \frac{\partial}{\partial r} \Phi}$$
$$- \frac{\Phi^2 \frac{\partial}{\partial r} P \frac{\partial^2}{(\partial t)^2} \Phi}{2 \left(\Phi^5 \frac{\partial}{\partial r} P - P \Phi^4 \frac{\partial}{\partial r} \Phi\right)} - \frac{P \Phi \frac{\partial}{\partial t} \Phi \frac{\partial^2}{\partial t \partial r} \Phi}{2 \left(\Phi^5 \frac{\partial}{\partial r} P - P \Phi^4 \frac{\partial}{\partial r} \Phi\right)} - \frac{P \Phi^2 \frac{\partial}{\partial r} \Phi}{2 \left(\Phi^5 \frac{\partial}{\partial r} P - P \Phi^4 \frac{\partial}{\partial r} \Phi\right)} - \frac{P \Phi \frac{\partial}{\partial t} \Phi \frac{\partial}{\partial r} \Phi}{\Phi^5 \frac{\partial}{\partial r} P - P \Phi^4 \frac{\partial}{\partial r} \Phi} - \frac{P \frac{\partial}{\partial t} \Phi^2 \frac{\partial}{\partial r} \Phi}{2 \left(\Phi^5 \frac{\partial}{\partial r} P - P \Phi^4 \frac{\partial}{\partial r} \Phi\right)} + \frac{P \Phi \frac{\partial^2}{(\partial t)^2} \Phi \frac{\partial}{\partial r} \Phi}{2 \left(\Phi^5 \frac{\partial}{\partial r} P - P \Phi^4 \frac{\partial}{\partial r} \Phi\right)}$$

$$
\begin{aligned}
\Lambda = & -\frac{P^2\Phi^5\frac{\partial^3}{(\partial x)^2\partial r}P\frac{\partial}{\partial r}P^2}{12\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}+\frac{P\Phi^5\frac{\partial}{\partial x}P\frac{\partial^2}{\partial x\partial r}P\frac{\partial}{\partial r}P^2}{6\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}-\frac{P^2\Phi^5\frac{\partial^3}{(\partial y)^2\partial r}P\frac{\partial}{\partial r}P^2}{12\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}\\
& +\frac{P\Phi^5\frac{\partial}{\partial y}P\frac{\partial^2}{\partial y\partial r}P\frac{\partial}{\partial r}P^2}{6\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}-\frac{\Phi^5\frac{\partial}{\partial x}P^2\frac{\partial}{\partial r}P^3}{4\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}+\frac{P\Phi^5\frac{\partial^2}{(\partial x)^2}P\frac{\partial}{\partial r}P^3}{6\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}\\
& -\frac{\Phi^5\frac{\partial}{\partial y}P^2\frac{\partial}{\partial r}P^3}{4\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}+\frac{P\Phi^5\frac{\partial^2}{(\partial y)^2}P\frac{\partial}{\partial r}P^3}{6\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}+\frac{P^3\Phi^4\frac{\partial^3}{(\partial x)^2\partial r}P\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi}{6\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}\\
& -\frac{P^2\Phi^4\frac{\partial}{\partial x}P\frac{\partial^2}{\partial x\partial r}P\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi}{3\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}+\frac{P^3\Phi^4\frac{\partial^3}{(\partial y)^2\partial r}P\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi}{6\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}-\frac{P^2\Phi^4\frac{\partial}{\partial y}P\frac{\partial^2}{\partial y\partial r}P\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi}{3\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}\\
& +\frac{7\,P\Phi^4\frac{\partial}{\partial x}P^2\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi}{12\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}-\frac{5\,P^2\Phi^4\frac{\partial^2}{(\partial x)^2}P\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi}{12\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}+\frac{7\,P\Phi^4\frac{\partial}{\partial y}P^2\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi}{12\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}\\
& -\frac{5\,P^2\Phi^4\frac{\partial^2}{(\partial y)^2}P\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi}{12\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}-\frac{P^4\Phi^3\frac{\partial^3}{(\partial x)^2\partial r}P\frac{\partial}{\partial r}\Phi^2}{12\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}+\frac{P^3\Phi^3\frac{\partial}{\partial x}P\frac{\partial^2}{\partial x\partial r}P\frac{\partial}{\partial r}\Phi^2}{6\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}\\
& -\frac{P^4\Phi^3\frac{\partial^3}{(\partial y)^2\partial r}P\frac{\partial}{\partial r}\Phi^2}{12\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}+\frac{P^3\Phi^3\frac{\partial}{\partial y}P\frac{\partial^2}{\partial y\partial r}P\frac{\partial}{\partial r}\Phi^2}{6\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}-\frac{5\,P^2\Phi^3\frac{\partial}{\partial x}P^2\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2}{12\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}\\
& +\frac{P^3\Phi^3\frac{\partial^2}{(\partial x)^2}P\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2}{3\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}-\frac{5\,P^2\Phi^3\frac{\partial}{\partial y}P^2\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2}{12\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}+\frac{P^3\Phi^3\frac{\partial^2}{(\partial y)^2}P\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2}{3\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}\\
& +\frac{P^3\Phi^2\frac{\partial}{\partial x}P^2\frac{\partial}{\partial r}\Phi^3}{12\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}-\frac{P^4\Phi^2\frac{\partial^2}{(\partial x)^2}P\frac{\partial}{\partial r}\Phi^3}{12\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}+\frac{P^3\Phi^2\frac{\partial}{\partial y}P^2\frac{\partial}{\partial r}\Phi^3}{12\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}\\
& -\frac{P^4\Phi^2\frac{\partial^2}{(\partial y)^2}P\frac{\partial}{\partial r}\Phi^3}{12\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}-\frac{\Phi^5\frac{\partial}{\partial r}P^3}{4\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}+\frac{\Phi^4\frac{\partial}{\partial r}P^3\frac{\partial^2}{(\partial t)^2}\Phi}{4\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}\\
& -\frac{P\Phi^4\frac{\partial}{\partial r}P^2\frac{\partial^3}{(\partial t)^2\partial r}\Phi}{12\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}-\frac{P\Phi^3\frac{\partial}{\partial r}P^2\frac{\partial}{\partial t}\Phi\frac{\partial^2}{\partial t\partial r}\Phi}{12\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}+\frac{2\,P\Phi^4\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi}{3\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}\\
& +\frac{P^2\Phi^4\frac{\partial^2}{(\partial r)^2}P\frac{\partial}{\partial r}\Phi}{12\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}+\frac{P\Phi^2\frac{\partial}{\partial r}P^2\frac{\partial}{\partial t}\Phi^2\frac{\partial}{\partial r}\Phi}{12\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}\\
& -\frac{2\,P\Phi^3\frac{\partial}{\partial r}P^2\frac{\partial^2}{(\partial t)^2}\Phi\frac{\partial}{\partial r}\Phi}{3\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}+\frac{P^2\Phi^3\frac{\partial}{\partial r}P\frac{\partial^3}{(\partial t)^2\partial r}\Phi\frac{\partial}{\partial r}\Phi}{6\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}+\frac{P^2\Phi^2\frac{\partial}{\partial r}P\frac{\partial}{\partial t}\Phi\frac{\partial^2}{\partial t\partial r}\Phi\frac{\partial}{\partial r}\Phi}{6\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}\\
& -\frac{2\,P^2\Phi^3\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2}{3\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}-\frac{P^2\Phi\frac{\partial}{\partial r}P\frac{\partial}{\partial t}\Phi^2\frac{\partial}{\partial r}\Phi^2}{6\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}+\frac{7\,P^2\Phi^2\frac{\partial}{\partial r}P\frac{\partial^2}{(\partial t)^2}\Phi\frac{\partial}{\partial r}\Phi^2}{12\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}\\
& -\frac{P^3\Phi^2\frac{\partial^3}{(\partial t)^2\partial r}\Phi\frac{\partial}{\partial r}\Phi^2}{12\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}-\frac{P^3\Phi\frac{\partial}{\partial t}\Phi\frac{\partial^2}{\partial t\partial r}\Phi\frac{\partial}{\partial r}\Phi^2}{12\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}+\frac{P^3\Phi^2\frac{\partial}{\partial r}\Phi^3}{4\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}\\
& +\frac{P^3\frac{\partial}{\partial t}\Phi^2\frac{\partial}{\partial r}\Phi^3}{12\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}-\frac{P^3\Phi\frac{\partial^2}{(\partial t)^2}\Phi\frac{\partial}{\partial r}\Phi^3}{6\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}-\frac{P^2\Phi^4\frac{\partial}{\partial r}P\frac{\partial^2}{(\partial r)^2}\Phi}{12\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}
\end{aligned}
$$

Weyl Tensor Components

$$\Psi_0 = -\frac{P^2 \frac{\partial^3}{(\partial x)^2 \partial r} P}{4\left(\Phi^2 \frac{\partial}{\partial r} P - P\Phi \frac{\partial}{\partial r}\Phi\right)} - \frac{i\, P^2 \frac{\partial^3}{\partial x \partial y \partial r} P}{2\left(\Phi^2 \frac{\partial}{\partial r} P - P\Phi \frac{\partial}{\partial r}\Phi\right)} + \frac{P^2 \frac{\partial^3}{(\partial y)^2 \partial r} P}{4\left(\Phi^2 \frac{\partial}{\partial r} P - P\Phi \frac{\partial}{\partial r}\Phi\right)} + \frac{P \frac{\partial^2}{(\partial x)^2} P \frac{\partial}{\partial r} P}{4\left(\Phi^2 \frac{\partial}{\partial r} P - P\Phi \frac{\partial}{\partial r}\Phi\right)} + \frac{i\, P \frac{\partial^2}{\partial x \partial y} P \frac{\partial}{\partial r} P}{2\left(\Phi^2 \frac{\partial}{\partial r} P - P\Phi \frac{\partial}{\partial r}\Phi\right)} - \frac{P \frac{\partial^2}{(\partial y)^2} P \frac{\partial}{\partial r} P}{4\left(\Phi^2 \frac{\partial}{\partial r} P - P\Phi \frac{\partial}{\partial r}\Phi\right)}$$

$$\Psi_1 = -\frac{P^2 \frac{\partial^2}{\partial x \partial r} P \frac{\partial}{\partial r}\Phi}{4\left(\Phi^3 \frac{\partial}{\partial r} P^2 - 2\, P\Phi^2 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi + P^2 \Phi \frac{\partial}{\partial r}\Phi^2\right)} - \frac{i\, P^2 \frac{\partial^2}{\partial y \partial r} P \frac{\partial}{\partial r}\Phi}{4\left(\Phi^3 \frac{\partial}{\partial r} P^2 - 2\, P\Phi^2 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi + P^2 \Phi \frac{\partial}{\partial r}\Phi^2\right)}$$
$$+ \frac{P \frac{\partial}{\partial x} P \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi}{4\left(\Phi^3 \frac{\partial}{\partial r} P^2 - 2\, P\Phi^2 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi + P^2 \Phi \frac{\partial}{\partial r}\Phi^2\right)} + \frac{i\, P \frac{\partial}{\partial y} P \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi}{4\left(\Phi^3 \frac{\partial}{\partial r} P^2 - 2\, P\Phi^2 \frac{\partial}{\partial r} P \frac{\partial}{\partial r}\Phi + P^2 \Phi \frac{\partial}{\partial r}\Phi^2\right)}$$

$$
\begin{aligned}
\Psi_2 = & -\frac{P^2\Phi^5\frac{\partial^3}{(\partial x)^2\partial r}P\frac{\partial}{\partial r}P^2}{12\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}+\frac{P\Phi^5\frac{\partial}{\partial x}P\frac{\partial^2}{\partial x\partial r}P\frac{\partial}{\partial r}P^2}{6\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}-\frac{P^2\Phi^5\frac{\partial^3}{(\partial y)^2\partial r}P\frac{\partial}{\partial r}P^2}{12\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}\\
& +\frac{P\Phi^5\frac{\partial}{\partial y}P\frac{\partial^2}{\partial y\partial r}P\frac{\partial}{\partial r}P^2}{6\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}-\frac{P\Phi^5\frac{\partial^2}{(\partial x)^2}P\frac{\partial}{\partial r}P^3}{12\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}-\frac{P\Phi^5\frac{\partial^2}{(\partial y)^2}P\frac{\partial}{\partial r}P^3}{12\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}\\
& +\frac{P^3\Phi^4\frac{\partial^3}{(\partial x)^2\partial r}P\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi}{6\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}-\frac{P^2\Phi^4\frac{\partial}{\partial x}P\frac{\partial^2}{\partial x\partial r}P\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi}{3\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}+\frac{P^3\Phi^4\frac{\partial^3}{(\partial y)^2\partial r}P\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi}{6\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}\\
& -\frac{P^2\Phi^4\frac{\partial}{\partial y}P\frac{\partial^2}{\partial y\partial r}P\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi}{3\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}-\frac{P\Phi^4\frac{\partial}{\partial x}P^2\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi}{6\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}+\frac{P^2\Phi^4\frac{\partial^2}{(\partial x)^2}P\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi}{3\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}\\
& -\frac{P\Phi^4\frac{\partial}{\partial y}P^2\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi}{6\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}+\frac{P^2\Phi^4\frac{\partial^2}{(\partial y)^2}P\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi}{3\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}-\frac{P^4\Phi^3\frac{\partial^3}{(\partial x)^2\partial r}P\frac{\partial}{\partial r}\Phi^2}{12\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}\\
& +\frac{P^3\Phi^3\frac{\partial}{\partial x}P\frac{\partial^2}{\partial x\partial r}P\frac{\partial}{\partial r}\Phi^2}{6\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}-\frac{P^4\Phi^3\frac{\partial^3}{(\partial y)^2\partial r}P\frac{\partial}{\partial r}\Phi^2}{12\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}+\frac{P^3\Phi^3\frac{\partial}{\partial y}P\frac{\partial^2}{\partial y\partial r}P\frac{\partial}{\partial r}\Phi^2}{6\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}\\
& +\frac{P^2\Phi^3\frac{\partial}{\partial x}P^2\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2}{3\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}-\frac{5\,P^3\Phi^3\frac{\partial^2}{(\partial x)^2}P\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2}{12\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}+\frac{P^2\Phi^3\frac{\partial}{\partial y}P^2\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2}{3\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}\\
& -\frac{5\,P^3\Phi^3\frac{\partial^2}{(\partial y)^2}P\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2}{12\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}-\frac{P^3\Phi^2\frac{\partial}{\partial x}P^2\frac{\partial}{\partial r}\Phi^3}{6\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}+\frac{P^4\Phi^2\frac{\partial^2}{(\partial x)^2}P\frac{\partial}{\partial r}\Phi^3}{6\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}\\
& -\frac{P^3\Phi^2\frac{\partial}{\partial y}P^2\frac{\partial}{\partial r}\Phi^3}{6\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}+\frac{P^4\Phi^2\frac{\partial^2}{(\partial y)^2}P\frac{\partial}{\partial r}\Phi^3}{6\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}+\frac{P\Phi^4\frac{\partial}{\partial r}P^2\frac{\partial^3}{(\partial t)^2\partial r}\Phi}{6\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}\\
& -\frac{P\Phi^3\frac{\partial}{\partial r}P^2\frac{\partial}{\partial t}\Phi\frac{\partial^2}{\partial t\partial r}\Phi}{3\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}+\frac{P\Phi^4\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi}{6\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}-\frac{P^2\Phi^4\frac{\partial^2}{(\partial r)^2}P\frac{\partial}{\partial r}\Phi}{6\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}\\
& +\frac{P\Phi^2\frac{\partial}{\partial r}P^2\frac{\partial}{\partial t}\Phi^2\frac{\partial}{\partial r}\Phi}{3\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}-\frac{P\Phi^3\frac{\partial}{\partial r}P^2\frac{\partial^2}{(\partial t)^2}\Phi\frac{\partial}{\partial r}\Phi}{6\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}-\frac{P^2\Phi^3\frac{\partial}{\partial r}P\frac{\partial^3}{(\partial t)^2\partial r}\Phi\frac{\partial}{\partial r}\Phi}{3\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}\\
& +\frac{2\,P^2\Phi^2\frac{\partial}{\partial r}P\frac{\partial}{\partial t}\Phi\frac{\partial^2}{\partial t\partial r}\Phi\frac{\partial}{\partial r}\Phi}{3\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}-\frac{P^2\Phi^3\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2}{6\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}-\frac{2\,P^2\Phi\frac{\partial}{\partial r}P\frac{\partial}{\partial t}\Phi^2\frac{\partial}{\partial r}\Phi^2}{3\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}\\
& +\frac{P^2\Phi^2\frac{\partial}{\partial r}P\frac{\partial^2}{(\partial t)^2}\Phi\frac{\partial}{\partial r}\Phi^2}{3\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}+\frac{P^3\Phi^2\frac{\partial^3}{(\partial t)^2\partial r}\Phi\frac{\partial}{\partial r}\Phi^2}{6\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}-\frac{P^3\Phi\frac{\partial}{\partial t}\Phi\frac{\partial^2}{\partial t\partial r}\Phi\frac{\partial}{\partial r}\Phi^2}{3\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}\\
& +\frac{P^3\frac{\partial}{\partial t}\Phi^2\frac{\partial}{\partial r}\Phi^3}{3\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}-\frac{P^3\Phi\frac{\partial^2}{(\partial t)^2}\Phi\frac{\partial}{\partial r}\Phi^3}{6\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}+\frac{P^2\Phi^4\frac{\partial}{\partial r}P\frac{\partial^2}{(\partial r)^2}\Phi}{6\left(\Phi^7\frac{\partial}{\partial r}P^3-3\,P\Phi^6\frac{\partial}{\partial r}P^2\frac{\partial}{\partial r}\Phi+3\,P^2\Phi^5\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi^2-P^3\Phi^4\frac{\partial}{\partial r}\Phi^3\right)}
\end{aligned}
$$

$$
\begin{aligned}
\Psi_3 = & \frac{P^2\frac{\partial^2}{\partial x\partial r}P\frac{\partial}{\partial r}\Phi}{4\left(\Phi^3\frac{\partial}{\partial r}P^2-2\,P\Phi^2\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi+P^2\Phi\frac{\partial}{\partial r}\Phi^2\right)}-\frac{i\,P^2\frac{\partial^2}{\partial y\partial r}P\frac{\partial}{\partial r}\Phi}{4\left(\Phi^3\frac{\partial}{\partial r}P^2-2\,P\Phi^2\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi+P^2\Phi\frac{\partial}{\partial r}\Phi^2\right)}\\
& -\frac{P\frac{\partial}{\partial x}P\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi}{4\left(\Phi^3\frac{\partial}{\partial r}P^2-2\,P\Phi^2\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi+P^2\Phi\frac{\partial}{\partial r}\Phi^2\right)}+\frac{i\,P\frac{\partial}{\partial y}P\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi}{4\left(\Phi^3\frac{\partial}{\partial r}P^2-2\,P\Phi^2\frac{\partial}{\partial r}P\frac{\partial}{\partial r}\Phi+P^2\Phi\frac{\partial}{\partial r}\Phi^2\right)}
\end{aligned}
$$

$$\Psi_4 = -\frac{P^2 \frac{\partial^3}{(\partial x)^2 \partial r} P}{4\left(\Phi^2 \frac{\partial}{\partial r} P - P\Phi \frac{\partial}{\partial r}\Phi\right)} + \frac{i\, P^2 \frac{\partial^3}{\partial x \partial y \partial r} P}{2\left(\Phi^2 \frac{\partial}{\partial r} P - P\Phi \frac{\partial}{\partial r}\Phi\right)} + \frac{P^2 \frac{\partial^3}{(\partial y)^2 \partial r} P}{4\left(\Phi^2 \frac{\partial}{\partial r} P - P\Phi \frac{\partial}{\partial r}\Phi\right)} + \frac{P \frac{\partial^2}{(\partial x)^2} P \frac{\partial}{\partial r} P}{4\left(\Phi^2 \frac{\partial}{\partial r} P - P\Phi \frac{\partial}{\partial r}\Phi\right)} - \frac{i\, P \frac{\partial^2}{\partial x \partial y} P \frac{\partial}{\partial r} P}{2\left(\Phi^2 \frac{\partial}{\partial r} P - P\Phi \frac{\partial}{\partial r}\Phi\right)} - \frac{P \frac{\partial^2}{(\partial y)^2} P \frac{\partial}{\partial r} P}{4\left(\Phi^2 \frac{\partial}{\partial r} P - P\Phi \frac{\partial}{\partial r}\Phi\right)}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "I"